

Climate Matters: Electric vehicles?rising sales and falling equities

Date: 2026-01-28

Source: <https://www.franklintempleton.com/articles/2025/equity/climate-matters-electric-vehicles-rising-sales-and-falling-equities>

Extraction method: selenium

Climate Matters: Electric vehicles?rising sales and falling equities

Having been strongly in favour during 2021 and 2022, stocks across the electric vehicle (EV) supply chain have struggled in recent years, as the previously infallible EV transition has failed to live up to expectations. Looking back, a favourable backdrop for stocks across the EV ecosystem gave rise to high valuations, often due to inflated supply chain pricing. In the last two years, valuations have declined significantly. Interest rates have risen, margins and pricing have come under pressure, growth expectations have been reduced, and earnings estimates have been revised down accordingly.

The news flow in the last two years has made for bleak reading for EVs. In the United States, executive orders from President Trump to remove EV incentives were announced on his first day in office. From Asia, 2024 brought images of vast fields of unsold electric cars on the outskirts of Hangzhou and other Chinese cities.¹ Meanwhile in Europe, there have been high-profile policy moves to water down incentives, such as the UK government?s decision to postpone by five years its target of banning new petrol car sales, and the EU?s decision to offer ?flexibility? around its 2025 carbon emissions targets for automotive manufacturers.

Perhaps this is no surprise, coming after Europe failed to grow EV demand at all in 2024. Reality fell short of expectations, and global EV growth slowed from 33% down to 24%.² A tilt to hybrids rather than ?full? electric also made this growth less impactful, both in terms of emissions reductions and on supply/demand implications in the value chain. Battery demand quickly fell below supply, and lithium prices collapsed by more than 80% from their 2022 highs. Capital expenditure adjustments have taken time to work through, and this ongoing reset has been difficult throughout the value chain.

The share price trajectories of companies involved in the EV transition is a stark reminder that growth generally needs to exceed or meet expectations for an investment thesis to work, particularly in more commoditised markets. Moreover, demand can soar without creating much shareholder value?a lesson the solar industry is also willing to teach anyone who is interested. Even at high growth rates, demand growth failing to match supply can create intense pricing pressure and margin declines.

Looking ahead, we have strong conviction that EV demand should continue to grow. But when evaluating the associated investment opportunities, we need to focus on where current expectations are and what we are paying for those equities. We do think the setup now is appealing?expectations have reset lower, and we believe valuations appear to be discounting limited growth.

We argue that from this starting point, the fundamental investment case underpinning most aspects of the EV supply chain?based upon continued technological advancements, regulatory support and growing environmental concerns?is solid. We hence look at several aspects of the EV supply chain where we believe attractive investment opportunities may be found?manufacturers of both EV batteries and semiconductors, as well as providers of the raw materials used in the EV production process.

Slowing growth ex-China creates supply chain opportunities

In our view, the auto sector is developing into two distinct regions progressing at different speeds ?China, now the world?s largest EV market, and global ex-China. China remains on track to exceed 70% EV adoption by 2030, having already achieved 49% EV penetration in Q4 2024.³ EV sales, including both fully battery-powered vehicles and hybrids, rose 39% in 2024 compared with 2023.⁴ This has been largely consumer driven as the Chinese market benefits from a wide range of affordable products.

The narrative of slowing EV demand is hence more a reflection of trends in Europe and the United States. Technology teething issues, receding subsidies, inflation, and relatively unaffordable price points for EVs vs traditional internal combustion varieties, have all contributed to a gloomy picture of faltering EV demand outside of China. Policy shifts have also pressured the space. Germany?s decision in late 2023 to end its EV subsidy program has been an important

driver of the recent slowdown, following cuts to subsidy programmes in France and the United Kingdom, leaving regulation and cost-competitiveness as the main levers to drive EV adoption.

Although a small number of countries in Asia are clearly most influential in EV production, the EV supply chain is truly global. It differs in several respects from the internal combustion engine (ICE) vehicle equivalent. EVs may be mechanically simpler and contain fewer parts, but the nature of the technology makes mass production quite challenging. Battery production, the most expensive component of these vehicles, is overwhelmingly concentrated in East Asia. The raw materials used in producing lithium-ion batteries, however, are scattered globally. Lithium is mined primarily in Australia and South America, nickel in Indonesia and the Philippines, and cobalt in central Africa. After mining and production, these materials are sent to China, South Korea or Japan for refinement, processing, and cathode and cell production before being sent on to the EV manufacturer. Power semiconductors are mostly produced in East Asia and the United States. In terms of actual EV manufacturing, China remains the clear leader, but Europe and the United States are also home to some leading producers.

So why focus on the supply chain? The reality is that for the automotive original equipment manufacturers (OEMs), the shift to EVs is difficult, costly, heavily regulated and technologically challenging. We believe that early winners, such as Tesla and BYD are clearly taking market share, but doing so by pressuring category margins in the medium-term and thus making the economics more difficult for the traditional OEMs. As such, the money to be made in the transition in our view lies not with the final product providers but those companies that supply to the EV industry. Many of these suppliers are enablers that help to lower the cost of EVs relative to traditional combustion vehicles or provide unique materials that EVs are particularly dependent upon.

Battery producers driving lower costs

Innovations to reduce battery costs are crucial in improving the affordability of EVs, as batteries are the primary source of cost differentiation. EV batteries contribute significantly to the overall vehicle price, even as costs have been declining. As many of the leading auto companies pursue plans to release newer, cost-effective models, we expect battery costs to fall further. This should have a major bearing on the viability and affordability of EVs generally, driving adoption. To enable EVs to achieve cost parity with ICE vehicles, increasing the energy density of batteries and optimizing the manufacturing process is important, and this is where makers of batteries make great partners for auto OEMs.

While some Chinese battery makers are leading in terms of scale and cost, we believe a leading South Korean battery manufacturer has successfully developed competitive advantages in battery chemistry, large scale manufacturing and supply chain integration. We also see promising early signs from a battery maker in India capitalising on domestic government-supported demand to the EV transition, that has also paired with automotive OEMs. Recent announcements by major cities to transition their public transportation fleets to entirely EVs should help spur a nascent EV battery industry in a country that is unlikely to rely on imported products.

Semiconductors crucial to electrification

Another enabler to drive EV adoption are power semiconductors, which help to convert battery power into motors throughout the vehicle. A leading manufacturer of semiconductors that we believe may benefit is now generating more than half of its revenues from power semiconductors, making it particularly exposed to EV adoption and industrial applications. Semiconductors and sensors in EVs are used to reduce power consumption and improve efficiency and safety. Notably, the semiconductor content in EV and hybrid vehicles is similar, so shifts in demand should be neutral, in our opinion. The products also have extensive uses in industrial applications, solar and wind farms, and consumer devices. While markets are fragmented, long duration contracts can average seven years in automotive, allowing time for companies to build integration and improve product efficiency. Switching costs are high, particularly in automotive, due to lengthy qualification times, mission critical applications, software lock-ins, and the low cost of chips relative to the price of the finished product.

Near-term market concerns regarding semiconductors relate to elevated inventories, weakness in EV sales, China softness and the ongoing correction in automotive and industrial end markets. Semiconductors stocks have underperformed for much of the period since mid-2023 on cycle concerns, and we think current levels offer an attractive entry point into what we see as a structural growth story driven by electrification.

Raw materials: Lithium

Moving further up the supply chain, several raw materials are particularly critical to the energy transition and electrification of vehicles. Lithium is a key driver of the decarbonization of the transportation industry, with the metal forming a major component of all EV batteries, regardless of cathode chemistry. More than 90% of lithium demand now comes from the battery industry,⁵ and EVs remain the main source of demand growth going forward.

High EV-battery inventory levels and weakening expectations around EV sales led to share-price weakness through 2023 and 2024, creating what we believe is a compelling long-term value opportunity in the lithium sector.

Since 2023, spot pricing for lithium carbonate at 99.5% concentration (lithium carbonate equivalent or LCE) has fallen from more than US\$75/kg to less than US\$10/kg. While we acknowledge that lithium prices could continue to be volatile in the near term, we believe the market is now pricing in the current low lithium price into perpetuity, a scenario where we see meaningful shortages in supply as EV demand grows.

Today significant quantities of lithium can be extracted at low cost from brines in the "lithium triangle" of Chile, Argentina and Bolivia, or from a suite of relatively new technologies known as direct lithium extraction (DLE). However, marginal sources of supply are still coming from non-integrated hard rock mining of a product known as spodumene. This precursor to lithium hydroxide and lithium carbonate is used in EV batteries after it is sold to conversion facilities to produce these lithium products. We see spodumene as a significant source of incremental supply as the EV market grows.

Our estimates suggest that prices are currently in the region of 40% too low to incentivize new supply of spodumene, and this discount is larger if alternative mined rocks are needed, such as lepidolite. Indeed, even the world's largest spodumene project, the Greenbushes mine in Australia, is planning to lower production in response to weak pricing. Yet, in our opinion, supply growth remains crucial in a market that is still expected to see demand grow at 10%+ through the rest of the decade.⁶ Notably we expect demand from China and Europe, where regulation and subsidies remain supportive, to drive this growth.

As EV demand growth continues and high levels of inventories unwind, we expect lithium prices to rebound at least to incentive pricing. We also see a meaningful chance of medium-term price spikes, similar to that experienced in 2022, on the back of ongoing price volatility. In such an environment, we expect lithium equities to do well against lowered expectations.

Raw materials: Aluminium

Transportation is also the largest end market for global aluminium, accounting for roughly 30% of total demand.⁷ The drive toward lighter vehicle construction has prompted increased demand for aluminium, with the silvery-white metal already replacing steel to some degree in traditional ICE vehicles. The light-weighting process can reduce transport emissions materially, on the simple concept that a lighter vehicle requires less power and thus less fuel.

Battery electric vehicles (BEVs) are generally 10%-15% heavier than their ICE counterparts, due to the weight of the battery pack and additional strengthening of the main structure to absorb increased crash loads.⁸ Thus, the need for light-weighting is greater with the move to EVs. Bernstein Research estimates that EVs use 250 kg of aluminium on average compared to 172 kg in ICEs,⁹ with the increase driven by the need to offset the weight of large batteries and battery enclosures. Aluminium's role in the light-weighting of the vehicle can also drive cost savings resulting from the ability to downsize the battery pack and drive-train components, while maintaining vehicle range.

In the face of increased regulations, there is enhanced focus on the CO₂ footprint of a vehicle, not just during the use phase, but also from a total-cycle assessment perspective. As part of this, both regulators and vehicle OEMs have sharpened their focus on the CO₂ footprint of raw materials used in the production process of the vehicle. The simple fact is that the average kilogram of aluminium produced globally today has a higher carbon footprint compared to the average kilogram of steel. However, given that aluminium is three times lighter, the comparison becomes less relevant as the weight savings compensates for the bulk of the difference.

The case for low-carbon aluminium is even more compelling as a superior alternative raw material for BEVs, as it allows for an emission reduction in both the production and use phases of the vehicle. This advantage starts at the first step of production, with automotive research firm *mfkare* reporting an average estimated CO₂ savings of around 15% vs steel by using low-carbon aluminium where practically possible in the production process. Amidst a backdrop of moderate oversupply in the aluminium sector, we particularly like a Norwegian producer of low-carbon aluminium that should stand to benefit from the increase in demand from both EVs but also for low-carbon aluminium specifically.

Looking ahead

While the recent landscape for the EV supply chain has plainly been challenging, we believe that many of the longer-term structural tailwinds for the theme remain intact. Slower growth, shifting policies, and excess inventories have tempered early enthusiasm, but they have also reset valuations and expectations to more realistic levels. This reset, in our view, presents an opportunity for investors with the correct approach. Rather than focusing on headline EV sales figures, we believe the greatest value lies in the enablers of electrification, including battery manufacturers, semiconductor producers, and raw material suppliers, where technological innovation and strategic positioning can still translate into attractive returns. As the EV transition matures, navigating it with a disciplined focus on fundamentals will likely be key to capturing sustainable value.

Endnotes

Source: "China's Abandoned Electric Cars Pile Up After EV Boom Fueled by Subsidies." Bloomberg. August 17, 2023.

Source: Bloomberg New Energy Finance "EV Market Outlook 1Q 2025."

Ibid.

Ibid.

Source: "Batteries and Secure Energy Transitions: World Energy Outlook Special 2024 Report." International Energy Agency (IEA). April 2024.

Source: BloombergNEF, Electric Vehicle Outlook.

Source: "Decarbonizing Aluminum: Rolling Out a More Sustainable Sector." Center for Strategic and International Studies (CSIS). February 25, 2022.

Source: "Global EV Outlook 2024." IEA.

Source: Ibid.

WHAT ARE THE RISKS?

All investments involve risks, including possible loss of principal.

Equity securities are subject to price fluctuation and possible loss of principal.

International investments are subject to special risks, including currency fluctuations and social, economic and political uncertainties, which could increase volatility. These risks are magnified in emerging markets. Investments in companies in a specific country or region may experience greater volatility than those that are more broadly diversified geographically.

The government's participation in the economy is still high and, therefore, investments in China will be subject to larger regulatory risk levels compared to many other countries.

There are special risks associated with investments in China, Hong Kong and Taiwan, including less liquidity, expropriation, confiscatory taxation, international trade tensions, nationalization, and exchange control regulations and rapid inflation, all of which can negatively impact the fund. Investments in Taiwan could be adversely affected by its political and economic relationship with China.

Any companies and/or case studies referenced herein are used solely for illustrative purposes; any investment may or may not be currently held by any portfolio advised by Franklin Templeton. The information provided is not a recommendation or individual investment advice for any particular security, strategy, or investment product and is not an indication of the trading intent of any Franklin Templeton managed portfolio.

WF: 5458761

CONTRIBUTORS

Craig Cameron, CFA

Portfolio Manager, Research Analyst, Templeton Global Investments

Tina M. Sadler, CFA

Portfolio Manager, Research AnalystTempleton Global Equity Group

Related Content

Explore Templeton Global Equity Group Insights

RELATED CONTENT

IMPORTANT LEGAL INFORMATION

This material is intended to be of general interest only and should not be construed as individual investment advice or a recommendation or solicitation to buy, sell or hold any security or to adopt any investment strategy. It does not constitute legal or tax advice. This material may not be reproduced, distributed or published without prior written permission from Franklin Templeton.

The views expressed are those of the investment manager and the comments, opinions and analyses are rendered as at publication date and may change without notice. The underlying assumptions and these views are subject to change based on market and other conditions and may differ from other portfolio managers or of the firm as a whole. The information provided in this material is not intended as a complete analysis of every material fact regarding any country, region or market. There is no assurance that any prediction, projection or forecast on the economy, stock market, bond market or the economic trends of the markets will be realized. The value of investments and the income from them can go down as well as up and you may not get back the full amount that you invested. Past performance is not necessarily indicative nor a guarantee of future performance. All investments involve risks, including possible loss of principal.

Any research and analysis contained in this material has been procured by Franklin Templeton for its own purposes and may be acted upon in that connection and, as such, is provided to you incidentally. Data from third party sources may have been used in the preparation of this material and Franklin Templeton ("FT") has not independently verified, validated or audited such data. Although information has been obtained from sources that Franklin Templeton believes to be reliable, no guarantee can be given as to its accuracy and such information may be incomplete or condensed and may be subject to change at any time without notice. The mention of any individual securities should neither constitute nor be construed as a recommendation to purchase, hold or sell any securities, and the information provided regarding such individual securities (if any) is not a sufficient basis upon which to make an investment decision. FT accepts no liability whatsoever for any loss arising from use of this information and reliance upon the comments, opinions and analyses in the material is at the sole discretion of the user.

Franklin Templeton has environmental, social and governance (ESG) capabilities; however, not all strategies or products for a strategy consider ESG as part of their investment process.

Products, services and information may not be available in all jurisdictions and are offered outside the U.S. by other FT affiliates and/or their distributors as local laws and regulation permits. Please consult your own financial professional or Franklin Templeton institutional contact for further information on availability of products and services in your jurisdiction.

Issued in the U.S. by Franklin Templeton, One Franklin Parkway, San Mateo, California 94403-1906, (800) DIAL BEN/342-5236, franklintempleton.com. Investments are not FDIC insured; may lose value; and are not bank guaranteed.

You need Adobe Acrobat Reader to view and print PDF documents. Download a free version from Adobe's website.

CFA® and Chartered Financial Analyst® are trademarks owned by CFA Institute.